Unit – V DESIGN THINKING :

Design thinking is a methodology that designers use to **brainstorm** and solve complex problems related to **designing and design engineering**. It is also beneficial for designers to find **innovative**, **desirable** and **never-thought-before** solutions for customers

Design thinking is used extensively in the area of **Healthcare and Wellness**, **Agriculture**, **Food security**, **Education**, **Financial services**, and **Environmental sustainability**, to name a few. Design thinking is a **blend of logic**, powerful imagination, systematic reasoning and **intuition to bring to the table** the ideas that promise to solve the problems of the clients with desirable outcomes. It helps to bring creativity with business insights.

The 5 Stages in the Design Thinking process:

The five stages of design thinking are:

Empathize: research your users' needs.
Define: state your users' needs and problems.
Ideate: challenge, assumptions and create ideas.
Prototype: start to create solutions.
Test: try your solutions out.

1.Empathize: research your users' needs :

The first stage of the design thinking process focuses on user-centric research. You want to gain an empathic understanding of the problem you are trying to solve. Consult experts to find out more about the area of concern and conduct observations to engage and empathize with your users

2**.Define**: state your users' needs and problems :

In the Define stage, you will organize the information you have gathered during the Empathize stage. You'll analyze your observations to define the core problems you and your team have identified up to this point. **Defining the problem and problem statement must be done in a human-centered manner.**

3.Ideate : challenge, assumptions and create ideas:

During the third stage of the design thinking process, designers are ready to generate ideas. You've grown to understand your users and their needs in the Empathize stage, and you've analyzed your observations in the Define stage to create a user centric problem statement. With this solid background, you and your team members can start to **look at the problem from different perspectives and ideate innovative solutions to your problem statement.**

4.**Prototype**: start to create solutions:

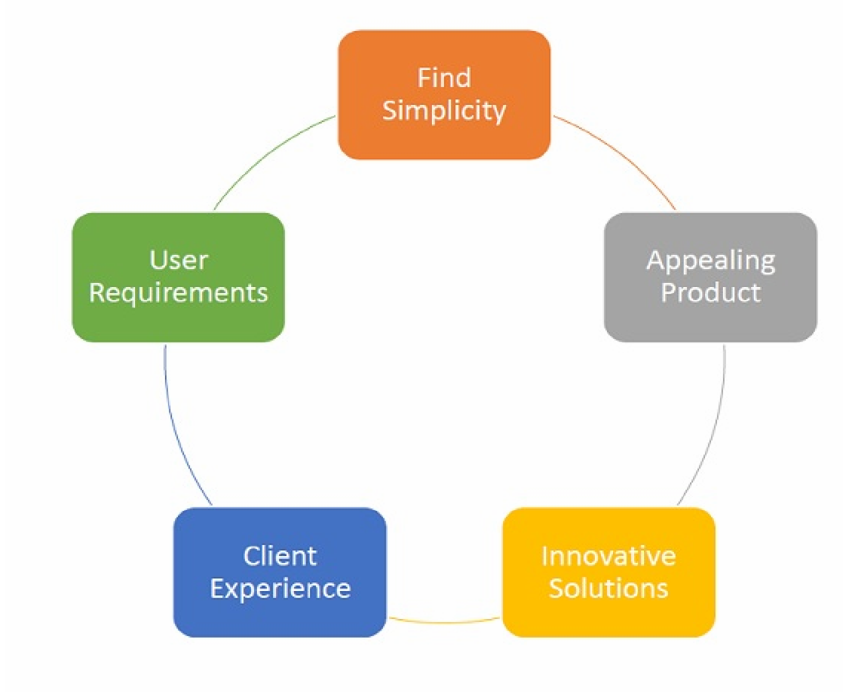
The design team will now produce a number of inexpensive, scaled down versions of the product (or specific features found within the product) to investigate the key solutions generated in the ideation phase. These prototypes can be shared and tested within the team itself, in other departments or on a small group of people outside the design team.

<u>**5.Test**: try your solutions out :</u>
Designers or evaluators rigorously test the complete product using the best solutions identified in the Prototype stage. This is the final stage of the five-stage model; however, in an iterative process such as design thinking, the results generated are often used to redefine one or more further problems.

Features of Design Thinking:

Such problems require multidimensional solutions. Design thinking helps in this regard. It not only assists a professional to come up with a solution, but it also helps the organization to gain a competitive edge over its rivals.

- Finding simplicity in complexities.
- Having a beautiful and aesthetically appealing product.
- Improving clients' and end user's quality of experience.
- Creating innovative, feasible, and viable solutions to real world problems.
- Addressing the actual requirements of the end users.



Design Thinking Applications :

Applications :

1.**<u>Business</u>**: Design thinking helps in businesses by **optimizing** the process of product creation, marketing, and renewal of contracts. All these processes require a companywide focus on the customer and hence, design thinking helps in these processes **immensely.**

2.**Information Technology**: The IT industry makes a lot of products that require **trials and proof of concepts**. The industry needs to empathize with its users and not simply deploy technologies. IT is not only about technology or products, but also its processes.

3.**Education**: The education sector can make the best use of design thinking by taking feedback from students on their requirements, goals and challenges they are facing in the classroom. By working on their feedback, the instructors can come up with solutions to address their challenges.

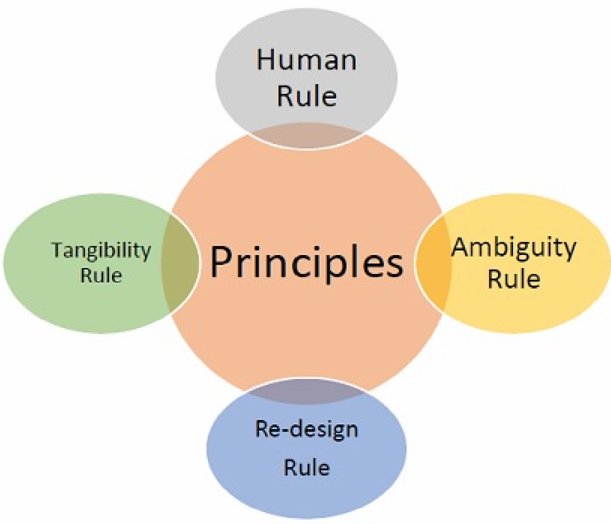
4.**HealthCare:** Design thinking helps in healthcare as well. The expenditure on healthcare by the government and the cost of healthcare facilities is growing by the day. Experts worldwide are concerned about how to bring quality healthcare to people at low cost.

Attributes in Design Thinking :

Design thinking is an extensive study of various attributes, like principles, methods and processes, challenges etc. Let's have a look at the attributes of design thinking.

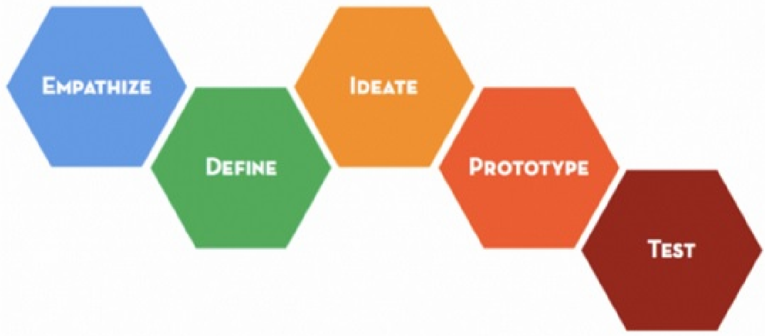
The Principles of Design Thinking

- **The Human Rule** – This rule states that all kinds of design activity are ultimately social in nature.
- **The Ambiguity Rule** – This rule requires all design thinkers to preserve ambiguity in the process design thinking.
- **The Re-design Rule** – The re-design rule states that all designs are basically examples of re-design.
- **The Tangibility Rule** – The tangibility rule states that making ideas tangible always facilitates communication between design thinkers.



The Five-step Process of Design Thinking :

The design thinking process or method has five steps in all to be followed. The process starts with **empathizing** with the problem of the customer or the end-user. The process then moves to **ideate** on solutions using divergent thinking. The prototype is developed after



convergent thinking and then the design thinkers resort to testing the prototype.

Lean Thinking

Lean thinking is a term used to describe the process of making business decisions in a Lean way. It’s regarded as the foundation of any Lean practice.

Principles of Lean Thinking:

Lean thinking can be broken down into these seven Lean principles that guide Lean thinking in businesses today. These principles are:

1. Optimize the whole
2. Eliminate waste
3. Create knowledge
4. Build quality in
5. Deliver fast by managing flow
6. Defer commitment
7. Respect people

Optimize the whole:

This principle acknowledges the need to take a holistic view of the software development process in order to make imporvements.

Eliminate waste:

The core idea of lean manufacturing is quite simple: **relentlessly** work on eliminating waste from the manufacturing process. What is waste? It can take many forms, the underlaying idea is to eliminating anything and everything that does not add value from the **perspective** of your customer. Anything that the customer would not willingly pay for is defined as waste in Lean thinking.

Create knowledge
The Lean thinking concept of Create Knowledge is simple: To scale, we must build learning and teaching into our organizations, so that more people can add more value in more ways. Of course, to do this, we must create environments that allow space for learning.

We create knowledge, and thereby improve, by taking the time to:

- Have **retrospectives**
- Hold team meetings to discuss how work in being done
- Cross-train team members

By creating knowledge throughout our organizations, we're able to deliver faster with more value.

Build quality in
To create a system that is built for growth, we need to error-proof our systems as much as possible. To do this, Lean organizations standardize and automate those things that are tedious, repeatable, or prone to human error, so they can focus the skills and efforts of their employees on innovation, growth, and continuous improvement.

Deliver fast by managing flow
At the core of Lean thinking is the idea that focus produces higher quality work. When our environment isn't designed to help us, focus, we move slower and slower than we would if we all had fewer things on our plates.

We must be more intentional about the way we work. Lean thinking encourages teams to visualize, manage, and continuously optimize their flow. If the goal is to deliver value to the customer at a sustainably fast pace, then we must have control over how work gets done; we must manage flow.

Defer commitment
Businesses often feel an artificial pressure to plan, make decisions, and complete work far in advance of when that specific value is needed by the market. This leads to a lack of flexibility that is necessary to continuously deliver value to customers.
The Lean thinking principle of Defer Commitment encourages organizations to make decisions at the last responsible moment, to continuously make decisions based on the most up-to-date, relevant, comprehensive information.

The reasoning behind this ties back to the concept of eliminating waste: If we plan and complete work before it's needed, then how do we know it will still be relevant to the market when we release it?

Respect people
Lean thinking reminds us that most of the value created in organizations is in the heads of employees, and that to retain those employees, we must create environments for employees to do their best work. Retaining quality talent is essential for sustainable value delivery.

The Lean interpretation of "respect" includes, of course, being a kind, courteous, and thoughtful employee, but it goes deeper than that. Lean organizations:

- Respect their employees by giving them what they need to do good work
- Create environments where the best ideas can be heard
- Encourage employees to pursue educational opportunities
- Give employees the autonomy to make decisions based on what is best for the customer

Actionable Strategy:

Design Thinking is in danger of becoming a methodology that people employ but don't understand. Design Thinking has become a more commonly used corporate phrase in the past few years. So much so that it has become a bit *buzzword*. Continuing in this vein will lead to it being implemented poorly and ultimately falling out of favour with decision-makers.

Though people understand the concept to a degree, the real benefits and purpose of Design Thinking can often be either misunderstood or completely opaque. This presents a particular challenge when trying to convince senior colleagues and those from non-creative roles that Design Thinking can play a vital role in tactical and strategic planning.

Innovation vs. Innovation Theatre

At this point, it is important to distinguish properly-run Design Thinking activities from "fluffy" design-themed box-ticking exercises these are often referred to as Innovation Theatre, because they are "just for show".

Clear Objectives Yield Good Outputs

The real value of Design Thinking exercises, activities, sprints or workshops comes from the outputs they deliver. -These outputs should always be clearly aligned to specific objectives. For example, if your objective is to determine why a certain element of your product or service has been performing poorly in a certain region or demographic, Design Thinking can help you drill down into your ecosystem, determine who the key users are, what their overall journey looks like, what their needs are at each of the steps in that journey, and what their motivations and frustrations are in relation to those needs.

Strategic Direction & Definition

The specific focus area to explore is strategy. By this, don't necessarily mean design strategy (how to approach or organise design) or design-led strategy (company-wide design centricity), but the overall strategy for a product, team or company.

For example, imagine a team that wants to outline a new strategic vision for a product category. The vision may include stretch goals, focus areas and action items.

It will outline some key activities that could be completed to deliver this strategy. This is by no means a comprehensive list or a specific process, but some examples of methodologies that can be used to good effect.

Discovery & Problem Definition

Developing a solution should always start with understanding the problem. In Design Thinking, you may come across *Problem Statements* or *Job Stories* to capture the key information. But whatever method or template you choose, this stage usually involves a lot of questions.

Brainstorming

This is many designers' bread and butter. Generate many possible solutions. +
disparate ideas. This may seem antithetical to strategic planning, but only when a large volume of ideas have been generated can a team effectively prioritise and determine best focus areas for a strategic direction.

The Problem with Complexity:→

A big challenge for many products and services is that they're deployed into highly uncertain and complex systems, like our economy, a customer ecosystem, or even a large enterprise. These complex adaptive systems are made up of interconnected but autonomous entities, acting and reacting to one another, without centralized control. Predicting a specific result or outcome in these conditions is incredibly difficult because behaviour in such a system is emergent. Although uncertainty is high, there are two characteristics we can be confident about in complex adaptive systems:

They're unknowable

There are multiple truths, and the system cannot be described from only one perspective or language. That is, nobody really understands everything. And we can't fully know anything.

They're intractable

That is, nobody is in control, the current condition emerges, and nothing is predictable.
This means that nobody knows what is going on, nobody can control it, and nobody can predict anything. Then there are humans—just one of the actors in the system—who are famously irrational in the way we behave.

Vision and Strategy:

What is a Vision?

A vision statement is a declaration of an organization's objectives. This statement can serve as a guide for the business, indicating a defined direction for growth and goals. As a company grows, its vision statement may change slightly to adapt to differing needs, but the main idea typically remains the same. Formal company documents—such as employee handbooks—usually contain the vision statement. However, some organizations advertise their statement on their website and other public platforms.

A vision statement should motivate the organization's employees and outline its ultimate goals. While a mission statement concentrates on what a company can provide for its customers, a vision statement is more self-focused, outlining the goals of the business for growth and success.

Consider the vision statement as an umbrella description of what is important to the company and how it helps to improve the world around it. This includes not only customers but potential employees as well. When the vision statement is able to connect to our inner values, it can be easy to admire that organization and support it in a variety of ways. We consider this a powerful communication tool in such a short statement.

An effective vision statement should be:

1. Clear to understand the organization's mission
2. Simple
3. Specific
4. Inspirational
5. Aware of challenges
6. Relevant to a variety of audiences
7. Authentic to the organization
8. Genuine
9. Unique from competitors

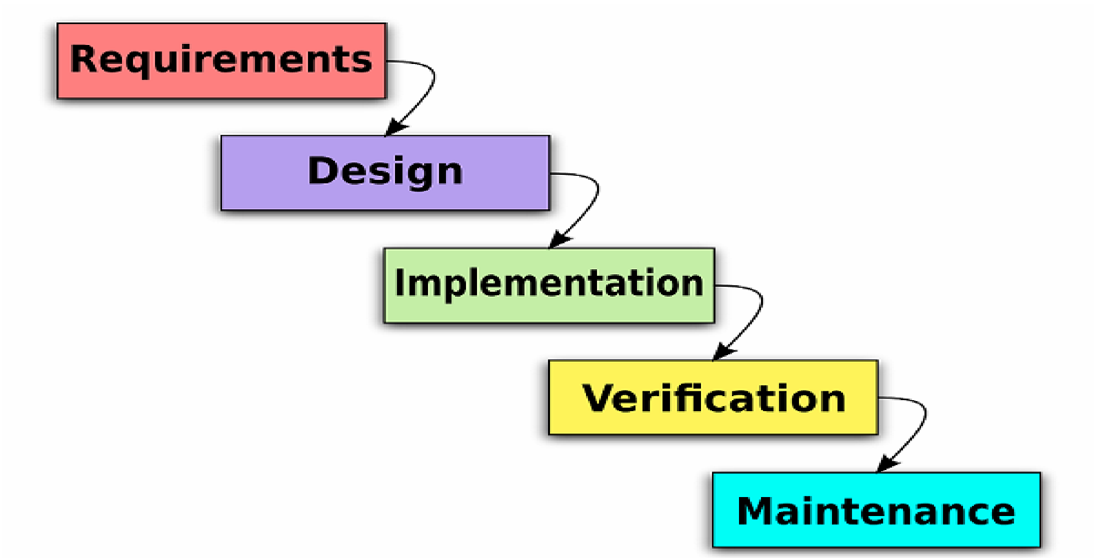
Drafting a motivational vision statement is an important step in identifying the goals of a business and how members of the organization can work toward accomplishing those objectives. It may be difficult to narrow multiple objectives into one-to-two sentences so it's important to really explore the strategy, products and services of the organization. It may require a marketing perspective or even a copywriter to help edit a larger plan or statement document.

Software development strategy.

1) Agile development strategy:

Most dedicated software development teams currently leverage agile scrum methodology in delivering their services. They believed in its effectiveness and efficiency, which benefits both clients and the team by accelerating the development process.

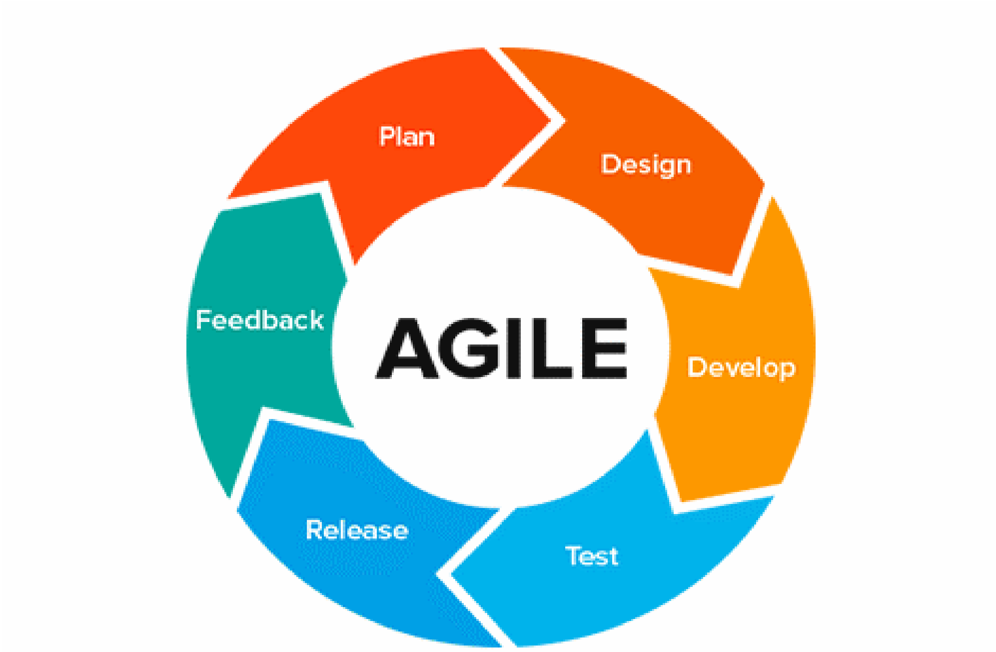
Waterfall



When the dedicated development team decides to apply the Waterfall strategy, they will strictly follow a detailed plan. Whereby every step has been clarified with no uncertainties. In Waterfall strategies, developers rarely change any minor part of the software development plan**,** that they follow step-by-step execution. Accordingly, the current task has to be done, before the next starting.

Unfortunately, the waterfall software development strategy currently is no longer popular that most dedicated developers believed it is outdate. However, it also owns undeniable advantages in software development, which admired the idea of equal priority among project parts.

2) Agile



Currently, agile software development as the dominant methodology has commonly applied in several offshore software outsourcing companies. Undeniable, Agile methodology brings immense benefits, but it requires the right direction in implementation.

For using an agile development strategy, you need a proficient team with skilled developers who indicate their technology proficiency. The concept of agile development strategy refers to the term agility or rapid software development. It means using Agile development strategy, experience and expertise are extremely important. Accordingly, the more skills developers have the faster and more undoubtedly the projects adopt.

With agile development strategy, you can steadily change or combine within three core strategies:

- Agile Scrum with separated phases
- Kanban with task board
- Informal conversation with instant messaging apps.

3) DevOps

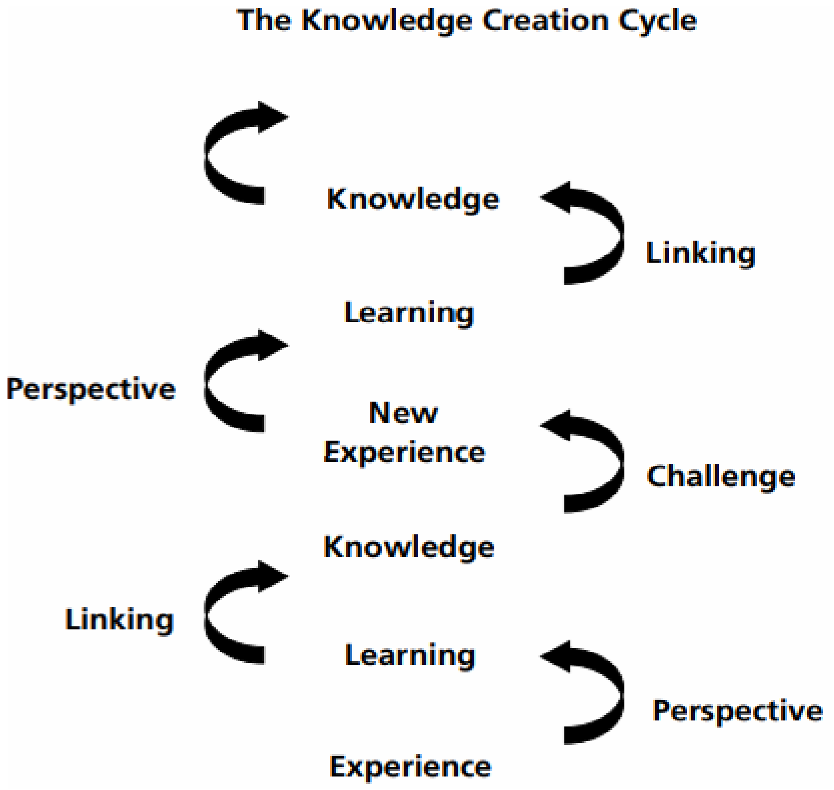
Besides interaction among IT infrastructure and dedicated developers, DevOps tend to be an advanced version of agile software development. DevOps was born to help developers break the technological limitation. In practice, leveraging DevOps helps developers in applying automation in the software development process, which bring tons of benefits:

- Rapid bug detects and fixing
- Improve software quality
- Cut down launching time
- Improve client satisfaction
- Speed up productivity and teamwork proficiency

But how can DevOps solve the entire above list? The DevOps backbone strategy focuses on individuals, who can be advanced in many positions and expertise. DevOps strategy admits the continuous release and process, which limit the painless for the development team.

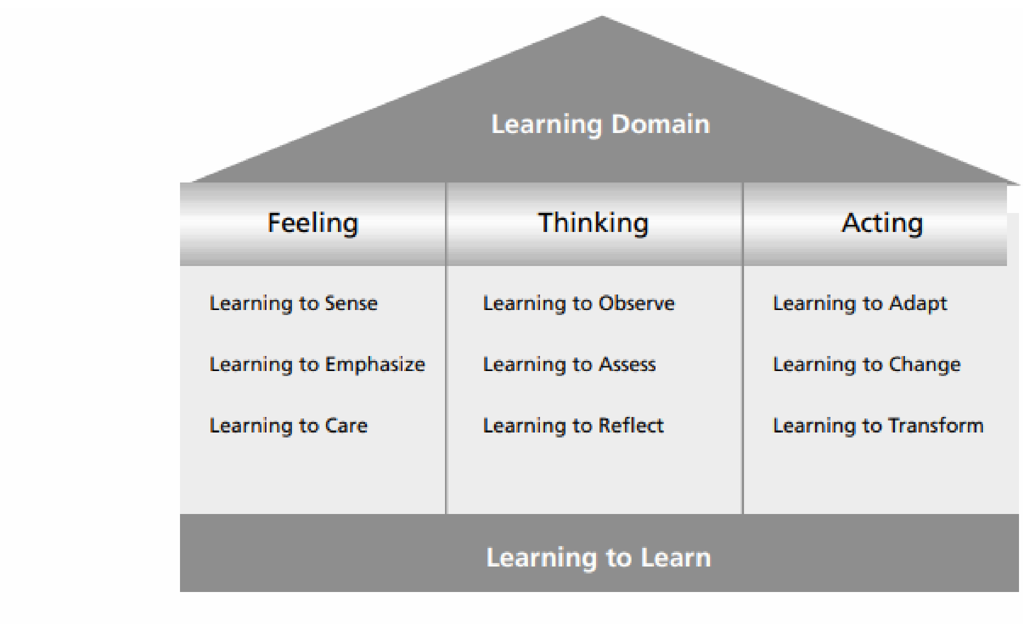
Defining Actionable Strategy Act To Learn Introduction:

Actionable Learning and the Knowledge Creation Cycle The actionable learning cycle provides contexts for and channels through which to create knowledge. Knowledge is often wrongly associated with command over information and facts. It is perceived to be "knowledge about" something, as if to have knowledge one must be at a distance from that which is to be known. Another way to view knowledge is to see it as existing on a continuum from the possession of factual information about something, through knowing how to do things in various contexts, to taking reflective action. This view of knowledge underpins the actionable learning framework for capacity building. By moving through the actionable learning framework, moreover, one is not only involved in learning, but also the creation of knowledge.



Framework:

The framework of actionable learning links three domains of learning: **emotion, thought** and **behaviour** , and underpins each with a growing capacity to learn how to learn. Specific learning capabilities are central to the capacity to learn within each domain. These capabilities are described briefly below.



Learning to Feel :

- **Learning to Sense** The capacity to be in touch with ones feeling and the feelings of others.
- **Learning to Empathize** The non-judgmental capacity to take the role of the other and examine ideas, situations and problems from that point of view.
- **Learning to Care** The critical capacity to be aware on ones values and the values of others and to elaborate and interrogate moral points of view.

Learning to Think :

- **Learning to Observe** The capacity to acquire, order, and seek patterns in empirical evidence of various forms.
- **Learning to Assess** The capacity to analyze information and draw defensible conclusions within various conceptual frameworks.
- **Learning to Reflect** The capacity to make and derive meaning from events, situations, myths, symbols and interactions.

Learning to Do:

- **Learning to Adapt** The capacity to shape and modify ones behavioural repertoire in light of changes in the environment.
- **Learning to Change** The capacity to undertake and deploy new behavioural repertoires in light of discontinuous change.
- **Learning to Transform** The continuous capacity to reinvent ones perceptual frames of reference and archetypal responses in light of ongoing changes and new experience.

Learning to Learn :

- The capacity to be conscious of personal modes of learning and to continually expand, differentiate and deepen the range and scope of one's learning capabilities.
- Actionable learning draws upon, and elaborates, all of these dimensions of learning and, in so doing, develops a person's capacity to learn how to feel, think, do and learn.
- Problems and challenges in capacity building, today, are not presented in a linear sequence, nor are they confined to issues of information and ideas alone.
- This point has been amply demonstrated in recent efforts to change and reform systems in developing countries.

- Many “good ideas”, presented crisply and confidently in conferences of experts, never see the light of day in the working world of people in developing countries, no matter what the force behind their prescription might be.
- The explanation for this is reasonably clear: new ideas, or even old ideas in new packages, require, for implementation, commitment and willpower.
- New ideas generate a host of reactions and impacts: they energize and disturb emotions and attachments, they demand new ways of acting, they perturb systems and make life uncomfortable for many, they call for change in what are often change aversive situations and most importantly they often call for new modes of learning.